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GRINDING TOOL DRESSING METHOD AND GRINDING DEVICE

Abstract

A method for dressing a grinding tool includes acquiring information on a deflection waveform indicating a positional deviation of a first tooth surface for each rotation phase of a first rotating body, acquiring, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the deflection waveform, the envelope passing through a plurality of crests of the deflection waveform located on a side closer to a second tooth surface, and dressing a grinding tooth surface with a dresser tooth surface by changing the rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-037060 filed on Mar. 11, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a grinding tool dressing method and a grinding device.

Description of the Related Art

[0003] JP 5367085 B2 discloses a method for dressing a grinding tool in which a grinding tool and a dresser gear are meshed and rotated, whereby a helical grinding tooth surface of the grinding tool is dressed by a dresser tooth surface of the dresser gear.

SUMMARY OF THE INVENTION

[0004] Better grinding tool dressing methods and grinding devices are long-awaited.

[0005] The present disclosure aims to solve the aforementioned problems.

[0006] A first aspect of the present disclosure is a grinding tool dressing method for dressing a helical grinding tooth surface of a grinding tool with a dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface or the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body including a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, the grinding tool dressing method comprising: acquiring information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body while a rotation speed of the first rotating body is changed in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed with each other and rotated, acquiring, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelop passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface, and dressing the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0007] A second aspect of the present disclosure is a grinding device configured to dress a helical grinding tooth surface of a grinding tool with the dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface and the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body having a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, and the grinding device comprising a rotation control unit configured to change a rotation speed of the first rotating body in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated, an information acquisition unit configured to acquire information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body, and a data acquisition

unit configured to acquire, as contact positional data indicating a position where the first tooth surface and the second tooth surface are contact with each other, an envelope of the deflection waveform acquired by the information acquisition unit, the envelope passing through a plurality of crests of the deflection waveform on a side closer to the second tooth surface, wherein the rotation control unit dresses the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0008] According to the present disclosure, a better grinding tool dressing method and grinding device can be provided.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of a grinding device according to an embodiment;

[0011] FIG. 2 is a control block diagram of the grinding device;

[0012] FIG. 3 is a flow chart illustrating one example of a grinding tool dressing method;

[0013] FIG. 4 is a flow chart showing an example of the grinding tool dressing method;

[0014] FIG. 5 is a graph showing deflection waveform of a left dresser tooth surface; and

[0015] FIG. 6 is a graph showing deflection waveform of a right dresser tooth surface.

DETAILED DESCRIPTION OF THE INVENTION

[0016] In a grinding tool dressing method, if an axial misalignment occurs when a dresser gear is mounted on a gear mounting shaft of a grinding device, the dresser gear vibrates during rotation. In this case, there is a case that a grinding tooth surface cannot be dressed by a dresser tooth surface.

[0017] As a method for solving such a problem, for example, the following grinding tool dressing method is considered. The grinding tool dressing method includes, for example, a first step, a second step, and a dressing step. In the first step, with the grinding tool and the dresser gear meshed and rotated, a deflection waveform that indicates a positional deviation of the dresser tooth surface for each rotation phase of the dresser gear is acquired while the rotation speed of the dresser gear is changed in a manner so that the grinding tooth surface and the dresser tooth surface come into contact with each other.

[0018] In the second step, a line passing through an average value of the positional deviations against the rotation phase of the dresser gear in the deflection waveform is acquired as reference positional data. In the second step, dressing positional data is acquired by adding a forming quantity (infeed data) to the reference positional data. In the dressing step, while the grinding tool and the dresser gear are meshed and rotated, the rotation speed of the dresser gear is changed based on the dressing positional data, whereby the grinding tooth surface is dressed by the dresser tooth surface. According to such a method, even when an axial misalignment occurs between, for example, the gear mounting shaft and the dresser gear, the grinding tooth surface can be dressed.

[0019] The dresser tooth surface has abrasive grains for dressing the grinding tooth surface. The abrasive grains are distributed in a nonuniform manner on the dresser tooth surface. The non-uniformity in the distribution of the abrasive grains on the dresser tooth surface appears as the inconsistencies in an amplitude of the deflection waveform. A circumferential speed of the dresser gear when the rotation speed of the dresser gear is changed increases as the outer diameter of the dresser gear increases. Therefore, the amplitude of the deflection waveform varies with the size and

the shape of the dresser gear.

[0020] A method of acquiring the dressing positional data by adding the forming quantity to the reference positional data passing through the average value for each rotation phase of the dresser gear in the deflection waveform may fail to accurately dress the grinding tooth surface due to the influence of the inconsistencies in the amplitude of the deflection waveform.

[0021] The present disclosure can provide a grinding tool dressing method and a grinding device capable of accurately dressing a grinding tooth surface without being affected by the inconsistencies in the amplitude of the deflection waveform.

[0022] FIG. 1 is a perspective view of a grinding device 10 according to an embodiment. As shown in FIG. 1, the grinding device 10 is a device for dressing a grinding tool 14 with a dresser gear 12. The grinding tool 14 can grind a workpiece gear (not shown) using the dressed grinding tool 14.

[0023] As shown in FIG. 1, the grinding device 10 includes a bed 16, a gear support mechanism 18, a gear rotation mechanism 20, a tool support mechanism 22, a tool rotation mechanism 24, and a controller 26.

[0024] The bed 16 is placed, for example, on a horizontal surface of a factory or the like. The gear support mechanism 18 is disposed on a flat upper surface of the bed 16. The gear support mechanism 18 includes a cutting table 28, a cutting motor 30, a traverse table 32, and a traverse motor 34.

[0025] The cutting table 28 moves in the A direction with respect to the bed 16. The A direction is a horizontal direction perpendicular to the height direction of the bed 16. The cutting table 28 is connected to the cutting motor 30 via a ball screw shaft 36. The cutting motor 30 moves the cutting table 28 in the A direction by rotating the ball screw shaft 36.

[0026] The traverse table 32 is disposed on the upper surface of the cutting table 28. The traverse table 32 moves in the B direction with respect to the cutting table 28. The B direction is a direction perpendicular to the height direction of the bed 16 and the A direction. The traverse table 32 is coupled to the traverse motor 34 via a ball screw shaft (not shown). The traverse motor 34 moves the traverse table 32 in the B direction by rotating the ball screw shaft.

[0027] The gear rotation mechanism 20 is arranged on an upper surface of the traverse table 32. The gear rotation mechanism 20 has a gear mounting shaft 38 and a first motor 40. The gear mounting shaft 38 extends in the B direction. The dresser gear 12 is attachable to and detachable from the gear mounting shaft 38. A work gear (not shown) can be mounted on the gear mounting shaft 38 in place of the dresser gear 12. The first motor 40 rotates the gear mounting shaft 38.

[0028] The tool support mechanism 22 includes a column 42, a pivot table 44, a shift table 46, and a shift motor 48. The column 42 is positioned on the upper surface of the bed 16 so as to face the gear support mechanism 18. The column 42 extends upward from the bed 16. The pivot table 44 is attached to a surface of the column 42 facing the gear support mechanism 18.

[0029] The pivot table 44 extends in one direction. A turning motor (not shown) turns the pivot table 44 in the C direction with respect to the column 42. The shift table 46 is provided on a surface of the pivot table 44 facing the gear support mechanism 18. The shift table 46 is coupled to the shift motor 48 via a ball screw shaft 50. The shift motor 48 is attached to the pivot table 44. The shift motor 48 moves the shift table 46 in the D direction with respect to the pivot table 44.

[0030] The tool rotation mechanism 24 includes a base 54, a tool mounting shaft 56, and a second motor 58. The base 54 is attached to a surface of the shift table 46 facing the gear support mechanism 18. The base 54 extends along the direction in which the pivot table 44 extends. The tool mounting shaft 56 is inserted through the base 54 along the direction in which the base 54 extends. The grinding tool 14 is attachable to and detachable from the tool mounting shaft 56. The second motor 58 rotates the tool mounting shaft 56.

[0031] As shown in FIG. 2, the dresser gear 12 is mounted on the gear mounting shaft 38. The dresser gear 12 can be rotated in the R1 direction and the R2 direction by the driving force of the first motor 40. The dresser gear 12 is a gear for dressing the grinding tool 14. The dresser gear 12

has a plurality of dresser teeth **60**. Each of the plurality of dresser teeth **60** is formed with a dresser tooth surface **62**. The dresser tooth surface **62** includes a left dresser tooth surface **62a** and a right dresser tooth surface **62b**. For example, diamond abrasive grains or the like are electrodeposited on the dresser tooth surface **62** via a nickel plating layer.

[0032] The grinding tool **14** is mounted on the tool mounting shaft **56**. The grinding tool **14** can be rotated in the R3 and R4 directions by the driving force of the second motor **58**. The grinding tool **14** is a tool for grinding a work gear (not shown). The grinding tool **14** has helical grinding teeth **64**. Grinding tooth surfaces **66** are formed on the grinding teeth **64**. The grinding tooth surface **66** includes a first grinding tooth surface **66a** and a second grinding tooth surface **66b**. For example, single-layer CBN (cubic boron nitride) abrasive grains or the like are electrodeposited on the grinding tooth surface **66** via a nickel plating layer.

[0033] When the grinding tool **14** is dressed by the dresser gear **12**, the dresser gear **12** and the grinding tool **14** are meshed with each other. With the dresser gear **12** and the grinding tool **14** meshed, the left dresser tooth surface **62a** faces the first grinding tooth surface **66a**, and the right dresser tooth surface **62b** faces the second grinding tooth surface **66b**. With the dresser gear **12** and the grinding tool **14** meshed with each other, the dresser gear **12** is rotated in the R1 direction and the grinding tool **14** is rotated in the R3 direction, whereby the first grinding tooth surface **66a** can be dressed by the left dresser tooth surface **62a**. With the dresser gear **12** and the grinding tool **14** meshed with each other, the dresser gear **12** is rotated in the R2 direction and the grinding tool **14** is rotated in the R4 direction, whereby the second grinding tooth surface **66b** can be dressed by the right dresser tooth surface **62b**.

[0034] The grinding device **10** further includes a first encoder **68**, a second encoder **70**, and a contact sensor **72**. The first motor **40** is provided with the first encoder **68**. The first encoder **68** outputs to the controller **26** information (e.g., pulse signals) on the rotation phase (rotation speed, rotation angle, rotation position, rotation amount) of the dresser gear **12**.

[0035] The second motor **58** is provided with the second encoder **70**. The second encoder **70** outputs to the controller **26** information (e.g., pulse signals) on the rotation phase (rotation speed, rotation angle, rotation position, and rotation amount) of the grinding tool **14**.

[0036] The contact sensor **72** detects the contact between the dresser tooth surface **62** and the grinding tooth surface **66**. The contact sensor **72** is mounted on a bearing (not shown) that supports the gear mounting shaft **38** in a rotatable manner. The contact sensor **72** is, for example, an AE (Acoustic Emission) sensor. The AE sensor detects an elastic wave (contact sound) generated when the dresser tooth surface **62** and the grinding tooth surface **66** come into contact with each other. The contact sensor **72** is not limited to the AE sensor. The contact sensor **72** may be, for example, a vibration sensor, a torque sensor, etc. The contact between the grinding tool **14** and the dresser gear **12** may be detected based on the accumulated pulse described in JP 3910427 B2.

[0037] The controller **26** includes a first servo amplifier **74**, a second servo amplifier **76**, and a control main body **78**. The first servo amplifier **74** controls the rotation of the first motor **40** based on signals output from the control main body **78**. The second servo amplifier **76** controls the rotation of the second motor **58** based on signals output from the control main body **78**.

[0038] The control main body **78** includes a computing unit **80**, a storage unit **82**, an operation unit **84**, and a display unit **86**. The computing unit **80** is composed of a processor such as a CPU (Central Processing Unit) or a GPU (Graphics Processing Unit). That is, the computing unit **80** is formed by processing circuitry.

[0039] The computing unit **80** includes a control unit **88**, a rotation control unit **90**, an information acquisition unit **92**, and a data acquisition unit **94**. The control unit **88** controls the cutting motor **30**, the traverse motor **34**, a turning motor (not shown), and the shift motor **48**. The rotation control unit **90** controls the rotation of the dresser gear **12** via the first servo amplifier **74**. The rotation control unit **90** controls the rotation of the grinding tool **14** via the second servo amplifier **76**.

[0040] The control unit **88**, the rotation control unit **90**, the information acquisition unit **92**, and the

data acquisition unit **94** can be realized by the computing unit **80** executing programs stored in the storage unit **82**. At least part of the control unit **88**, the rotation control unit **90**, the information acquisition unit **92**, and the data acquisition unit **94** may be realized by an integrated circuit such as an ASIC (Application Specific Integrated Circuit), an FPGA (Field-Programmable Gate Array), or the like. In addition, at least part of the control unit **88**, the rotation control unit **90**, the information acquisition unit **92**, and the data acquisition unit **94** may be configured by an electronic circuit including discrete devices.

[0041] The storage unit **82** is composed of a volatile memory (not shown) and a nonvolatile memory (not shown). Examples of the volatile memory include, for example, a RAM (Random Access Memory) or the like. The volatile memory is used as working memory of a processor to temporarily store data or the like required for processing or computing operations. Examples of the nonvolatile memory include, for example, a ROM (Read Only Memory), a flash memory, or the like. The non-volatile memory is used as memory for storage, storing programs, tables, maps, etc. At least part of the storage unit **82** may be provided in the above-described processor, integrated circuit, etc.

[0042] The operation unit **84** is used when a user operates the controller **26**. The operation unit **84** may include a keyboard, a mouse, and the like. The display unit **86** is provided with a display element (not shown). As the display element, for example, a liquid crystal display element, an organic electroluminescence display element, or the like is used. The operation unit **84** and the display unit **86** may be configured by a touch panel (not shown) provided with such a display element.

[0043] Next, an example of a method for dressing the grinding tool **14** will be described. FIGS. 3 and 4 are flow charts illustrating one example of a method for dressing the grinding tool **14**.

[0044] In step S1, the dresser gear **12** is mounted on the gear mounting shaft **38** and the grinding tool **14** is mounted on the tool mounting shaft **56**. Thereafter, the process transitions to step S2.

[0045] In step S2, the dresser gear **12** and the grinding tool **14** are meshed with each other. Specifically, the control unit **88** controls the cutting motor **30**, the traverse motor **34**, the turning motor (not shown), and the shift motor **48** to mesh the dresser gear **12** and the grinding tool **14** with each other. Thereafter, the process transitions to step S3.

[0046] In step S3, dressing positional data **104a** (see FIG. 5) of the first grinding tooth surface **66a** is acquired. That is, in step S10 of FIG. 4, an information acquisition step is performed. In the information acquisition step, in a state where the dresser gear **12** (first rotating body **110**) and the grinding tool **14** (second rotating body **114**) are meshed with each other and rotated, information on a deflection waveform **100a** (see FIG. 5) indicating the positional deviation of the left dresser tooth surface **62a** for each rotation phase of the dresser gear **12** is acquired while the rotation speed of the dresser gear **12** is changed in a manner so that the left dresser tooth surface **62a** (first tooth surface **112**) and the first grinding tooth surface **66a** (second tooth surface **116**) are in contact with each other.

[0047] Specifically, the rotation control unit **90** controls the first motor **40** via the first servo amplifier **74** to rotate the dresser gear **12** in the R1 direction. The rotation control unit **90** controls the second motor **58** via the second servo amplifier **76** to rotate the grinding tool **14** in the R3 direction. The rotation control unit **90** controls the first servo amplifier **74** and the second servo amplifier **76** using feedback in a manner so that the dresser gear **12** and the grinding tool **14** rotate maintaining the engaged state, based on the information output from the first encoder **68** and the information output from the second encoder **70**.

[0048] For example, when an axial misalignment occurs between the dresser gear **12** and the gear mounting shaft **38**, the left dresser tooth surface **62a** of the dresser gear **12** does not stably contact the first grinding tooth surface **66a** of the grinding tool **14**. In other words, when the dresser gear **12** and the grinding tool **14** are meshed and rotated, the left dresser tooth surface **62a** is in contact with and separated from the first grinding tooth surface **66a**. Therefore, in this embodiment, the

rotation control unit **90** changes the rotation speed of the dresser gear **12** in a manner so that the left dresser tooth surface **62a** and the first grinding tooth surface **66a** come into contact with each other. The rotation control unit **90** rotates the grinding tool **14** at a predetermined constant rotation speed. [0049] The contact sensor **72** detects elastic waves generated by the left dresser tooth surface **62a** and the first grinding tooth surface **66a** coming into contact with each other and outputs the detected elastic waves to the control main body **78**. The control unit **88** determines whether the magnitude of the elastic waves is within a predetermined contact range. For example, the lower limit of the contact range can be set to a value slightly larger than elastic waves in a state where the left dresser tooth surface **62a** and the first grinding tooth surface **66a** are not in contact with each other. For example, the upper limit of the contact range can be set to a value slightly smaller than the elastic waves in a state where the left dresser tooth surface **62a** is excessively in contact with the first grinding tooth surface **66a**. That is, if the magnitude of the elastic waves output from the contact sensor **72** is within the contact range, the left dresser tooth surface **62a** and the first grinding tooth surface **66a** are in contact with each other in an appropriate manner. The contact range can be set as appropriate.

[0050] When the control unit **88** determines that the magnitude of the elastic waves output from the contact sensor **72** is smaller than the lower limit of the contact range, the rotation control unit **90** increases the rotation speed of the dresser gear **12**. Thus, the left dresser tooth surface **62a** moves in the direction towards the first grinding tooth surface **66a**, so that the left dresser tooth surface **62a** and the first grinding tooth surface **66a** can be brought into contact with each other in an appropriate manner.

[0051] When the control unit **88** determines that the magnitude of the elastic waves output from the contact sensor **72** is larger than the upper limit value of the contact range, the rotation control unit **90** reduces the rotation speed of the dresser gear **12**. Thus, the left dresser tooth surface **62a** moves in the direction away from the first grinding tooth surface **66a**, so that the left dresser tooth surface **62a** and the first grinding tooth surface **66a** can be brought into contact with each other in an appropriate manner.

[0052] Thus, when the rotational speed of the dresser gear **12** is changed in a manner so that the left dresser tooth surface **62a** and the first grinding tooth surface **66a** come into contact with each other, the deflection of the left dresser tooth surface **62a** is reflected in the information output from the first encoder **68**. Therefore, the information acquisition unit **92** can acquire information on the deflection waveform **100a** indicating the positional deviation of the dresser tooth surface **62** for each rotation phase of the dresser gear **12**, based on the information output from the first encoder **68**.

[0053] FIG. 5 is a graph showing the deflection waveform **100a** of the left dresser tooth surface **62a**. In the graph of FIG. 5, the horizontal axis indicates the rotation phase of the dresser gear **12**, and the vertical axis indicates a positional deviation amount of the left dresser tooth surface **62a**. In this embodiment, the information acquisition unit **92** acquires information on the deflection waveform **100a** over one rotation of the dresser gear **12**. After this, the rotation of the dresser gear **12** and the grinding tool **14** is temporarily stopped, and the process shifts to step S11.

[0054] In step S11, a data acquisition step is performed. In the data acquisition step, as shown in FIG. 5, an envelope passing through multiple crests on a side of the deflection waveform **100a** acquired in the information acquisition step, the side being closer to the first grinding tooth surface **66a**, are acquired as contact positional data **102a** indicating positions at which the left dresser tooth surface **62a** and the first grinding tooth surface **66a** are in contact with each other.

[0055] In the data acquisition step, the rotational phase of the dresser gear **12** may be divided into a plurality of sections (e.g., **128** sections) in the deflection waveform **100a**; the positional deviation amount at the position where the phase is the smallest may be extracted as a representative point from each section; and the contact positional data **102a** may be acquired based on these representative points. In this case, since the load of data processing is reduced, the contact

positional data **102a** can be acquired in a relatively short time.

[0056] In the data acquisition step, dressing positional data **104a** is acquired in which a predetermined forming quantity is added to the contact positional data **102a**. Specifically, the contact positional data **102a** shown in FIG. 5 is slid toward a side closer to the first grinding tooth surface **66a** by the forming quantity, whereby the dressing positional data **104a** of the first grinding tooth surface **66a** is acquired. The forming quantity is machining allowance for the abrasive grains of the first grinding tooth surface **66a** and is set as appropriate. The forming quantity is stored in advance in the storage unit **82**. After this, the process transitions to step S4 (dressing step) in FIG. 3.

[0057] In step S4, the first grinding tooth surface **66a** is dressed. Specifically, in a state where the dresser gear **12** and the grinding tool **14** are meshed and rotated, the rotation speed of the dresser gear **12** is changed based on the dressing positional data **104a** of the first grinding tooth surface **66a**, whereby the first grinding tooth surface **66a** is dressed by the left dresser tooth surface **62a**. The dressing of the first grinding tooth surface **66a** is performed over the circumference of the dresser gear **12**. Thereafter, the process transitions to step S5.

[0058] In step S5, dressing positional data **104b** (see FIG. 6) of the second grinding tooth surface **66b** is acquired. The dressing positional data **104b** of the second grinding tooth surface **66b** may be acquired by performing the processing of steps S10 and S11 described above between the right dresser tooth surface **62b** and the second grinding tooth surface **66b**. The description of the portion overlapping with the description of the acquisition of the dressing positional data **104a** of the first grinding tooth surface **66a** is omitted.

[0059] That is, in this case, in step S10, in the information acquisition step, in a state where the dresser gear **12** and the grinding tool **14** are meshed with each other and rotated, the information on the deflection waveform **100b** (see FIG. 6) indicating a positional deviation of the right dresser tooth surface **62b** for each rotation phase of the dresser gear **12** is acquired while the rotation speed of the dresser gear **12** is changed in a manner so that the right dresser tooth surface **62b** (first tooth surface **112**) and the second grinding tooth surface **66b** (second tooth surface **116**) are in contact with each other. FIG. 6 is a graph showing the deflection waveform of the right dresser tooth surface **62b**.

[0060] In step S11, as shown in FIG. 6, an envelope passing through multiple crests on a side of the deflection waveform **100b** of the right dresser tooth surface **62b**, the side closer to the second grinding tooth surface **66b**, are acquired as the contact positional data **102b** indicating positions at which the right dresser tooth surface **62b** and the second grinding tooth surface **66b** are in contact with each other. Further, the contact positional data **102b** shown in FIG. 6 is slid toward a side closer to the second grinding tooth surface **66b** by the forming quantity, whereby the dressing positional data **104b** of the second grinding tooth surface **66b** is obtained. The forming quantity in this case is machining allowance for the abrasive grains of the second grinding tooth surface **66b** and is set as appropriate. Then, the process transitions to step S6 (dressing step).

[0061] In step S6, the second grinding tooth surface **66b** is dressed. Specifically, with the dresser gear **12** and the grinding tool **14** meshed and rotated, the rotation speed of the dresser gear **12** is changed based on the dressing positional data **104b** of the second grinding tooth surface **66b**, whereby the second grinding tooth surface **66b** is dressed by the right dresser tooth surface **62b**. The dressing of the second grinding tooth surface **66b** is performed over the circumference of the dresser gear **12**. After this, the process of FIG. 3 is completed.

[0062] In this embodiment, the dressing positional data **104a**, **104b** need not be acquired after the contact positional data **102a**, **102b** are acquired in the data acquisition step. In this case, in the dressing step, for example, the rotation control unit **90** may change the rotation speed of the dresser gear **12** by separately referring to the contact positional data **102a**, **102b** and the forming quantity.

[0063] According to the present embodiment, the envelopes passing through the multiple crests on sides of the deflection waveforms **100a**, **100b** of the dresser tooth surface **62**, the sides closer to the

grinding tooth surface **66**, are acquired as the contact positional data **102a**, **102b** indicating positions where the dresser tooth surface **62** and the grinding tooth surface **66** are in contact with each other. Thus, the contact positional data **102a**, **102b** (reference position) between the dresser tooth surface **62** and the grinding tooth surface **66** can be grasped with high accuracy. That is, the contact positional data **102a**, **102b** are not affected by the inconsistencies in the amplitude of the deflection waveforms **100a**, **100b**. The dresser gear **12** is changed based on the predetermined forming quantity and the contact positional data **102a**, **102b**, whereby the grinding tooth surface **66** is dressed by the dresser tooth surface **62**. Thus, the grinding tooth surface **66** can be dressed with high accuracy without being affected by the inconsistencies in the amplitude of the deflection waveforms **100a**, **100b**. Thus, a better method for dressing the grinding tool **14** and grinding device **10** can be provided.

[0064] In the above-described embodiment, the dresser gear **12** is a first rotating body **110** having a first tooth surface **112**, which is the dresser tooth surface **62**. The grinding tool **14** is a second rotating body **114** having a second tooth surface **116**, which is the grinding tooth surface **66**. The present disclosure is not limited to such a configuration, and for example, the grinding tool **14** may be the first rotating body **110** and the dresser gear **12** may be the second rotating body **114**. In this case, in the information acquisition step, in a state where the grinding tool **14** (first rotating body **110**) and the dresser gear **12** (second rotating body **114**) are meshed with each other and rotated, information on the deflection waveforms **100a**, **100b** indicating the positional deviation of the grinding tooth surface **66** for each rotation phase of the grinding tool **14** is acquired while the rotation speed of the grinding tool **14** is changed in a manner so that the grinding tooth surface **66** (first tooth surface **112**) and the dresser tooth surface **62** (second tooth surface **116**) come into contact with each other. In the data acquisition step, the envelopes passing through the multiple crests on the sides of the deflection waveforms **100a**, **100b** acquired in the information acquisition step, the sides closer to the grinding tooth surface **66**, are acquired as the contact positional data **102a**, **102b** indicating positions the dresser tooth surface **62** and the grinding tooth surface **66** are in contact with each other. Furthermore, in the dressing step, while the dresser gear **12** and the grinding tool **14** are meshed and rotated, the rotation speed of the grinding tool **14** is changed based on the predetermined forming quantity and the contact positional data **102a**, **102b**, whereby the grinding tooth surface **66** is dressed by the dresser tooth surface **62**.

[0065] With respect to the above embodiments, the following supplementary notes are further disclosed.

(Supplementary Note 1)

[0066] A method of the present disclosure for dressing a grinding tool (**14**) is a grinding tool dressing method for dressing a helical grinding tooth surface (**66**) of the grinding tool with a dresser tooth surface (**62**) of a dresser gear (**12**), wherein one of the grinding tool or the dresser gear is a first rotating body (**110**) including a first tooth surface (**112**) that is one of the grinding tooth surface or the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body (**114**) including a second tooth surface (**116**) that is the other of the grinding tooth surface or the dresser tooth surface, the grinding tool dressing method comprising: acquiring information on a deflection waveform (**100a**, **100b**) indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body while a rotation speed of the first rotating body is changed in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed with each other and rotated, acquiring, as contact positional data (**102a**, **102b**) indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelope passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface, and dressing the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state

where the first rotating body and the second rotating body are meshed and rotated.

[0067] According to such a method, an envelope passing through multiple crests located on a side closer to the second tooth surface among the deflection waveform of the first tooth surface are acquired as the contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other. Thus, the contact positional data (reference position) between the first and second tooth surfaces can be grasped with high accuracy. That is, the contact positional data are not affected by the inconsistencies in the amplitude of the deflection waveforms. The first gear is changed based on the predetermined forming quantity and the contact positional data, whereby the grinding tooth surface is dressed by the dresser tooth surface. Thus, the grinding tooth surface can be dressed with high accuracy without being affected by the inconsistencies in the amplitude of the deflection waveforms. Thus, a better method for dressing the grinding tool can be provided.

(Supplementary Note 2)

[0068] Regarding the grinding tool dressing method according to Supplementary note 1, the first tooth surface of the first rotating body may be the dresser tooth surface of the dresser gear, and the second tooth surface of the second rotating body may be the grinding tooth surface of the grinding tool.

(Supplementary Note 3)

[0069] Regarding the grinding tool dressing method according to Supplementary note 1 or 2, in the acquiring of the envelop, dressing positional data (**104a**, **104b**) is acquired in which the forming quantity is added to the contact positional data; and in the dressing, the grinding tooth surface is dressed with the dresser tooth surface by changing a rotation speed of the first rotating body based on the dressing positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0070] According to such a method, it is possible to dress the grinding tooth surface with the dresser tooth surface by simple control by acquiring the dressing positional data.

(Supplementary Note 4)

[0071] A grinding device of the present disclosure is a grinding device (**10**) configured to dress a helical grinding tooth surface of a grinding tool with the dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface and the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body having a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, and the grinding device comprising a rotation control unit (**90**) configured to change a rotation speed of the first rotating body in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated, an information acquisition unit (**92**) configured to acquire information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body, and a data acquisition unit (**94**) configured to acquire, as contact positional data indicating a position where the first tooth surface and the second tooth surface are contact with each other, an envelope of the deflection waveform acquired by the information acquisition unit, the envelope passing through a plurality of crests of the deflection waveform on a side closer to the second tooth surface, wherein the rotation control unit dresses the grinding tooth surface with the dresser tooth surface by changing a rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0072] According to such a configuration, a grinding device that achieves the same effect as that of Supplementary note 1 can be obtained. Thus, a better grinding device can be provided.

(Supplementary Note 5)

[0073] Regarding the grinding device according to Supplementary note 4, the first tooth surface of

the first rotating body may be the dresser tooth surface of the dresser gear, and the second tooth surface of the second rotating body may be the grinding tooth surface of the grinding tool.

(Supplementary Note 6)

[0074] Regarding the grinding device according to supplementary note 4 or 5, the data acquisition unit may acquire dressing positional data in which the forming quantity is added to the contact positional data, and the rotation control unit may dress the grinding tooth surface by the dresser tooth surface by changing the rotation speed of the first rotating body based on the dressing positional data in a state where the first rotating body and the second rotating body are meshed and rotated.

[0075] Although the present disclosure has been detailed, the present disclosure is not limited to the individual embodiments described above. These embodiments may be variously added, replaced, altered, partially deleted, etc., without departing from the scope of the present disclosure or the intent of the present disclosure as derived from the claims and their equivalents. These embodiments can also be implemented in combination. For example, in the above-described embodiment, the order of the operations and the order of the processes are shown as an example, and are not limited to these. The same applies to the case where numerical values or mathematical expressions are used in the description of the above-described embodiment.

Claims

1. A grinding tool dressing method for dressing a helical grinding tooth surface of a grinding tool with a dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface or the dresser tooth surface, another of the grinding tool or the dresser gear is a second rotating body including a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, the grinding tool dressing method comprising: acquiring information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body while a rotation speed of the first rotating body is changed in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated; acquiring, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelope passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface; and dressing the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in the state where the first rotating body and the second rotating body are meshed and rotated.
2. The grinding tool dressing method according to claim 1, wherein the first tooth surface of the first rotating body is the dresser tooth surface of the dresser gear, and the second tooth surface of the second rotating body is the grinding tooth surface of the grinding tool.
3. The grinding tool dressing method according to claim 1, further comprising: acquiring dressing positional data in which the forming quantity is added to the contact positional data; and dressing the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on the dressing positional data in the state where the first rotating body and the second rotating body are meshed and rotated.
4. A grinding device configured to dress a helical grinding tooth surface of a grinding tool with a dresser tooth surface of a dresser gear, wherein one of the grinding tool or the dresser gear is a first rotating body including a first tooth surface that is one of the grinding tooth surface or the dresser tooth surface, the other of the grinding tool or the dresser gear is a second rotating body including a second tooth surface that is the other of the grinding tooth surface or the dresser tooth surface, the

grinding device comprising a controller including one or more processors executing computer-executable instructions stored in memory, wherein the one or more processors execute the computer-executable instructions to cause the controller to: change a rotation speed of the first rotating body in a manner so that the first tooth surface and the second tooth surface are in contact with each other in a state where the first rotating body and the second rotating body are meshed and rotated; acquire information on a deflection waveform indicating a positional deviation of the first tooth surface for each rotation phase of the first rotating body; acquire, as contact positional data indicating a position where the first tooth surface and the second tooth surface are in contact with each other, an envelope of the acquired deflection waveform, the envelop passing through a plurality of crests of the deflection waveform located on a side closer to the second tooth surface, and dress the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on a predetermined forming quantity and the contact positional data in the state where the first rotating body and the second rotating body are meshed and rotated.

5. The grinding device according to claim 4, wherein the first tooth surface of the first rotating body is the dresser tooth surface of the dresser gear, and the second tooth surface of the second rotating body is the grinding tooth surface of the grinding tool.

6. The grinding device according to claim 4, wherein the one or more processors execute the computer-executable instructions to cause the controller to: acquire dressing positional data in which the forming quantity is added to the contact positional data; and dress the grinding tooth surface with the dresser tooth surface by changing the rotation speed of the first rotating body based on the dressing positional data in the state where the first rotating body and the second rotating body are meshed and rotated.
